

DATABASE MANAGEMENT SYSTEM LAB

ASSIGNMENT NO. 09 – [Name : Ridafatema Patel, Rollno. 63]

Step 1 : Starting mongo db

iot@iot-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~\$ mongosh

Current Mongosh Log ID: 69d625a3c19618fbaf44ba88

Connecting to:

mongodb://127.0.0.1:27017/?

directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.8.2

Using MongoDB: 7.0.31

Using Mongosh: 2.8.2

For mongosh info see: <https://www.mongodb.com/docs/mongodb-shell/>

The server generated these startup warnings when booting

2026-04-08T13:25:49.904+05:30: Using the XFS filesystem is strongly recommended with the WiredTiger storage engine. See

<http://dochub.mongodb.org/core/prodnotes-filesystem>

2026-04-08T13:25:52.594+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted

test> use RIDA;
switched to db RIDA

1. Create a collection named Book.

RIDA> db.CreateCollection(Book)

ReferenceError: Book is not defined

RIDA> db.CreateCollection("Book")

TypeError: db.CreateCollection is not a function

RIDA> db.createCollection("Book")

{ ok: 1 }

2. Insert 5 records

RIDA>

db.Book.insertMany([{Title:"DBMS",Description:"Database Management System",By:"Avhad Mam",URL:"kkw.edu.in",Tags:"sql",Likes:"50"},
{},{},{},{}

{
acknowledged: true,
insertedIds: {
'0': ObjectId('69d629dcc19618fbaf44ba89'),
'1': ObjectId('69d629dcc19618fbaf44ba8a'),
'2': ObjectId('69d629dcc19618fbaf44ba8b'),
'3': ObjectId('69d629dcc19618fbaf44ba8c'),
'4': ObjectId('69d629dcc19618fbaf44ba8d')

RIDA>

db.Book.insertMany([{Title:"DBMS",Description:"Database Management System",By:"Avhad Mam",URL:"kkw.edu.in",Tags:"sql",Likes:"50"},
{Title:"OOP",Description:"OOP Concepts",By:"Kalyani Mam",URL:"kkw.edu.in",Tags:"oop",Likes:"20"},
{Title:"DS",Description:"DataStructure & Algorithms",By:"Khaire Mam",URL:"kkw.edu.in",Tags:"ds",Likes:"38"}])

{
acknowledged: true,
insertedIds: {
'0': ObjectId('69d62ac9c19618fbaf44ba8e'),
'1': ObjectId('69d62ac9c19618fbaf44ba8f'),
'2': ObjectId('69d62ac9c19618fbaf44ba90')
}
}

-Displaying records

RIDA> db.Book.find()

[
{
_id: ObjectId('69d629dcc19618fbaf44ba89'),
Title: 'DBMS',
Description: 'Database Management System',
By: 'Avhad Mam',
URL: 'kkw.edu.in',
Tags: 'sql',
Likes: '50'
},
{_id: ObjectId('69d629dcc19618fbaf44ba8a') },
{_id: ObjectId('69d629dcc19618fbaf44ba8b') },
{_id: ObjectId('69d629dcc19618fbaf44ba8c') },
{_id: ObjectId('69d629dcc19618fbaf44ba8d') },
{
_id: ObjectId('69d62ac9c19618fbaf44ba8e'),
Title: 'DBMS',
Description: 'Database Management System',
By: 'Avhad Mam',
URL: 'kkw.edu.in',
Tags: 'sql',
Likes: '50'
},
{
_id: ObjectId('69d62ac9c19618fbaf44ba8f'),
Title: 'OOP',
Description: 'OOP Concepts',
By: 'Kalyani Mam',
URL: 'kkw.edu.in',
Tags: 'oop',
Likes: '20'

```

    },
    {
      _id: ObjectId('69d62ac9c19618fbaf44ba90'),
      Title: 'DS',
      Description: 'DataStructure & Algorithms',
      By: 'Khaire Mam',
      URL: 'kkw.edu.in',
      Tags: 'ds',
      Likes: '38'
    }
  ]
}
3. Insert 1 more record in collection
RIDA>
db.Book.insertOne([ {Title:"DCN",Description:"Data Communication and Networking",By:"Bairagi Mam",
URL:"kkw.edu.in",Tags:"dcn",Likes:"29",Username:"ridafatema29",Comments:"Excellent"} ])
{
  acknowledged: true,
  insertedId: ObjectId('69d62c01c19618fbaf44ba91')
}
RIDA> db.Book.find()
[
  {
    _id: ObjectId('69d629dcc19618fbaf44ba89'),
    Title: 'DBMS',
    Description: 'Database Management System',
    By: 'Avhad Mam',
    URL: 'kkw.edu.in',
    Tags: 'sql',
    Likes: '50'
  },
  { _id: ObjectId('69d629dcc19618fbaf44ba8a') },
  { _id: ObjectId('69d629dcc19618fbaf44ba8b') },
  { _id: ObjectId('69d629dcc19618fbaf44ba8c') },
  { _id: ObjectId('69d629dcc19618fbaf44ba8d') },
  {
    _id: ObjectId('69d62ac9c19618fbaf44ba8e'),
    Title: 'DBMS',
    Description: 'Database Management System',
    By: 'Avhad Mam',
    URL: 'kkw.edu.in',
    Tags: 'sql',
    Likes: '50'
  },
  {
    _id: ObjectId('69d62ac9c19618fbaf44ba8f'),
    Title: 'OOP',
    Description: 'OOP Concepts',
    By: 'Kalyani Mam',
    URL: 'kkw.edu.in',
    Tags: 'oop',
    Likes: '20'
  },
  {

```

```

    {
      _id: ObjectId('69d62ac9c19618fbaf44ba90'),
      Title: 'DS',
      Description: 'DataStructure & Algorithms',
      By: 'Khaire Mam',
      URL: 'kkw.edu.in',
      Tags: 'ds',
      Likes: '38'
    },
    {
      '0': {
        Title: 'DCN',
        Description: 'Data Communication and Networking',
        By: 'Bairagi Mam',
        URL: 'kkw.edu.in',
        Tags: 'dcn',
        Likes: '29',
        Username: 'ridafatema29',
        Comments: 'Excellent'
      },
      _id: ObjectId('69d62c01c19618fbaf44ba91')
    }
  ]
}
4. Display all records whose title is DCN & DBMS
RIDA> db.Book.find({Title:"DCN"}) // not fetched
RIDA> db.Book.find({Title:"DBMS"})
[
  {
    _id: ObjectId('69d629dcc19618fbaf44ba89'),
    Title: 'DBMS',
    Description: 'Database Management System',
    By: 'Avhad Mam',
    URL: 'kkw.edu.in',
    Tags: 'sql',
    Likes: '50'
  },
  {
    _id: ObjectId('69d62ac9c19618fbaf44ba8e'),
    Title: 'DBMS',
    Description: 'Database Management System',
    By: 'Avhad Mam',
    URL: 'kkw.edu.in',
    Tags: 'sql',
    Likes: '50'
  }
]
RIDA> db.Book.find({$or:[{By:"Kalyani Mam"},{Title:"OOP"}]})
[
  {
    _id: ObjectId('69d62ac9c19618fbaf44ba8f'),
    Title: 'OOP',
    Description: 'OOP Concepts',
    By: 'Kalyani Mam',

```

```

URL: 'kkw.edu.in',
Tags: 'oop',
Likes: '20'
}
]

```

5. or condition

```

RIDA> db.Book.find({$or:[{By:"Kalyani Mam"},
{Title:"DS"}]})

```

```

[
{
  _id: ObjectId('69d62ac9c19618fbaf44ba8f'),
  Title: 'OOP',
  Description: 'OOP Concepts',
  By: 'Kalyani Mam',
  URL: 'kkw.edu.in',
  Tags: 'oop',
  Likes: '20'
},
{
  _id: ObjectId('69d62ac9c19618fbaf44ba90'),
  Title: 'DS',
  Description: 'DataStructure & Algorithms',
  By: 'Khaire Mam',
  URL: 'kkw.edu.in',
  Tags: 'ds',
  Likes: '38'
}
]

```

6. and condition

```

RIDA> db.Book.find({$and:[{By:"Kalyani Mam"},
{Title:"DS"}]})

```

```

RIDA> db.Book.find({$and:[{By:"Kalyani Mam"},
{Title:"OOP"}]})

```

```

[
{
  _id: ObjectId('69d62ac9c19618fbaf44ba8f'),
  Title: 'OOP',
  Description: 'OOP Concepts',
  By: 'Kalyani Mam',
  URL: 'kkw.edu.in',
  Tags: 'oop',
  Likes: '20'
}
]

```

7. Greater than condition

```

RIDA> db.Book.find({Likes:{$gt:"25"}})

```

```

[
{
  _id: ObjectId('69d629dcc19618fbaf44ba89'),
  Title: 'DBMS',
  Description: 'Database Management System',
  By: 'Avhad Mam',
  URL: 'kkw.edu.in',
  Tags: 'sql',
  Likes: '50'
}
]

```

```

},
{
  _id: ObjectId('69d62ac9c19618fbaf44ba8e'),
  Title: 'DBMS',
  Description: 'Database Management System',
  By: 'Avhad Mam',
  URL: 'kkw.edu.in',
  Tags: 'sql',
  Likes: '50'
},
{
  _id: ObjectId('69d62ac9c19618fbaf44ba90'),
  Title: 'DS',
  Description: 'DataStructure & Algorithms',
  By: 'Khaire Mam',
  URL: 'kkw.edu.in',
  Tags: 'ds',
  Likes: '38'
}
]

```

8. Less than condition

```

RIDA> db.Book.find({Likes:{$lt:"30"}})

```

```

[
{
  _id: ObjectId('69d62ac9c19618fbaf44ba8f'),
  Title: 'OOP',
  Description: 'OOP Concepts',
  By: 'Kalyani Mam',
  URL: 'kkw.edu.in',
  Tags: 'oop',
  Likes: '20'
}
]

```

```

RIDA> db.Book.find({Likes:{$lt:"30"},$or:
[{Title:"OOP"},{By:"Khaire Mam"}]})

```

```

[
{
  _id: ObjectId('69d62ac9c19618fbaf44ba8f'),
  Title: 'OOP',
  Description: 'OOP Concepts',
  By: 'Kalyani Mam',
  URL: 'kkw.edu.in',
  Tags: 'oop',
  Likes: '20'
}
]

```

```

RIDA> db.Book.find({Likes:{$gt:"50"},$or:
[{Title:"OOP"},{By:"Khaire Mam"}]})

```

```

RIDA> db.Book.find({Likes:{$gt:"40"},$or:
[{Title:"OOP"},{By:"Khaire Mam"}]})

```

```

RIDA> db.Book.find({Likes:{$gt:"30"},$or:
[{Title:"OOP"},{By:"Khaire Mam"}]})

```

```

[
{
  _id: ObjectId('69d62ac9c19618fbaf44ba90'),
}
]

```

```

    Title: 'DS',
    Description: 'DataStructure & Algorithms',
    By: 'Khaire Mam',
    URL: 'kkw.edu.in',
    Tags: 'ds',
    Likes: '38'
  }
]
9. update the title DS to DSA
RIDA> db.Book.updateMany({Title:"DS"},{$set:
{Title:"DSA"}});
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
RIDA> show collections
Book
RIDA> db.Book.find()
[
  {
    _id: ObjectId('69d629dcc19618fbaf44ba89'),
    Title: 'DBMS',
    Description: 'Database Management System',
    By: 'Avhad Mam',
    URL: 'kkw.edu.in',
    Tags: 'sql',
    Likes: '50'
  },
  { _id: ObjectId('69d629dcc19618fbaf44ba8a') },
  { _id: ObjectId('69d629dcc19618fbaf44ba8b') },
  { _id: ObjectId('69d629dcc19618fbaf44ba8c') },
  { _id: ObjectId('69d629dcc19618fbaf44ba8d') },
  {
    _id: ObjectId('69d62ac9c19618fbaf44ba8e'),
    Title: 'DBMS',
    Description: 'Database Management System',
    By: 'Avhad Mam',
    URL: 'kkw.edu.in',
    Tags: 'sql',
    Likes: '50'
  },
  {
    _id: ObjectId('69d62ac9c19618fbaf44ba8f'),
    Title: 'OOP',
    Description: 'OOP Concepts',
    By: 'Kalyani Mam',
    URL: 'kkw.edu.in',
    Tags: 'oop',
    Likes: '20'
  },
  {
    _id: ObjectId('69d62ac9c19618fbaf44ba90'),

```

```

    Title: 'DSA',
    Description: 'DataStructure & Algorithms',
    By: 'Khaire Mam',
    URL: 'kkw.edu.in',
    Tags: 'ds',
    Likes: '38'
  },
  {
    '0': {
      Title: 'DCN',
      Description: 'Data Communication and
Networking',
      By: 'Bairagi Mam',
      URL: 'kkw.edu.in',
      Tags: 'dcn',
      Likes: '29',
      Username: 'ridafatema29',
      Comments: 'Excellent'
    },
    _id: ObjectId('69d62c01c19618fbaf44ba91')
  }
]
10. Delete the record titled 'DSA'
RIDA> db.Book.deleteMany({Title:"DSA"});
{ acknowledged: true, deletedCount: 1 }
RIDA> db.Book.find()
[
  {
    _id: ObjectId('69d629dcc19618fbaf44ba89'),
    Title: 'DBMS',
    Description: 'Database Management System',
    By: 'Avhad Mam',
    URL: 'kkw.edu.in',
    Tags: 'sql',
    Likes: '50'
  },
  { _id: ObjectId('69d629dcc19618fbaf44ba8a') },
  { _id: ObjectId('69d629dcc19618fbaf44ba8b') },
  { _id: ObjectId('69d629dcc19618fbaf44ba8c') },
  { _id: ObjectId('69d629dcc19618fbaf44ba8d') },
  {
    _id: ObjectId('69d62ac9c19618fbaf44ba8e'),
    Title: 'DBMS',
    Description: 'Database Management System',
    By: 'Avhad Mam',
    URL: 'kkw.edu.in',
    Tags: 'sql',
    Likes: '50'
  },
  {
    _id: ObjectId('69d62ac9c19618fbaf44ba8f'),
    Title: 'OOP',
    Description: 'OOP Concepts',
    By: 'Kalyani Mam',
    URL: 'kkw.edu.in',

```

```

    Tags: 'oop',
    Likes: '20'
  },
  {
    '0': {
      Title: 'DCN',
      Description: 'Data Communication and
Networking',
      By: 'Bairagi Mam',
      URL: 'kkw.edu.in',
      Tags: 'dcn',
      Likes: '29',
      Username: 'ridafatema29',
      Comments: 'Excellent'
    },
    _id: ObjectId('69d62c01c19618fbaf44ba91')
  }
]

```

11. & 12. display exactly 1 record of Avhad mam after skiping first

```

RIDA>db.Book.find({By:"Avhad
Mam"}).skip(1).limit(1);

```

```

[
  {
    _id: ObjectId('69d62ac9c19618fbaf44ba8e'),
    Title: 'DBMS',
    Description: 'Database Management System',
    By: 'Avhad Mam',
    URL: 'kkw.edu.in',
    Tags: 'sql',
    Likes: '50'
  }
]

```

```

RIDA>db.Book.find({By:"Kalyani
Mam"}).skip(1).limit(1);

```

13. Display all the books in sorted fashion

```

RIDA> db.Book.find().sort({Title :1});

```

```

[
  { _id: ObjectId('69d629dcc19618fbaf44ba8a') },
  { _id: ObjectId('69d629dcc19618fbaf44ba8b') },
  { _id: ObjectId('69d629dcc19618fbaf44ba8c') },
  { _id: ObjectId('69d629dcc19618fbaf44ba8d') },
  {
    '0': {
      Title: 'DCN',
      Description: 'Data Communication and
Networking',
      By: 'Bairagi Mam',
      URL: 'kkw.edu.in',
      Tags: 'dcn',
      Likes: '29',
      Username: 'ridafatema29',
      Comments: 'Excellent'
    },

```

```

    _id: ObjectId('69d62c01c19618fbaf44ba91')
  },
  {
    _id: ObjectId('69d629dcc19618fbaf44ba89'),
    Title: 'DBMS',
    Description: 'Database Management System',
    By: 'Avhad Mam',
    URL: 'kkw.edu.in',
    Tags: 'sql',
    Likes: '50'
  },
  {
    _id: ObjectId('69d62ac9c19618fbaf44ba8e'),
    Title: 'DBMS',
    Description: 'Database Management System',
    By: 'Avhad Mam',
    URL: 'kkw.edu.in',
    Tags: 'sql',
    Likes: '50'
  },
  {
    _id: ObjectId('69d62ac9c19618fbaf44ba8f'),
    Title: 'OOP',
    Description: 'OOP Concepts',
    By: 'Kalyani Mam',
    URL: 'kkw.edu.in',
    Tags: 'oop',
    Likes: '20'
  }
]

```

-Display all the books in desc fashion

```

RIDA> db.Book.find().sort({Title :-1});

```

```

[
  {
    _id: ObjectId('69d62ac9c19618fbaf44ba8f'),
    Title: 'OOP',
    Description: 'OOP Concepts',
    By: 'Kalyani Mam',
    URL: 'kkw.edu.in',
    Tags: 'oop',
    Likes: '20'
  },
  {
    _id: ObjectId('69d629dcc19618fbaf44ba89'),
    Title: 'DBMS',
    Description: 'Database Management System',
    By: 'Avhad Mam',
    URL: 'kkw.edu.in',
    Tags: 'sql',
    Likes: '50'
  },
  {
    _id: ObjectId('69d62ac9c19618fbaf44ba8e'),
    Title: 'DBMS',
    Description: 'Database Management System',

```

```
By: 'Avhad Mam',
URL: 'kkw.edu.in',
Tags: 'sql',
Likes: '50'
},
{ _id: ObjectId('69d629dcc19618fbaf44ba8a') },
{ _id: ObjectId('69d629dcc19618fbaf44ba8b') },
{ _id: ObjectId('69d629dcc19618fbaf44ba8c') },
{ _id: ObjectId('69d629dcc19618fbaf44ba8d') },
{
  '0': {
    Title: 'DCN',
    Description: 'Data Communion and
Networking',
    By: 'Bairagi Mam',
    URL: 'kkw.edu.in',
    Tags: 'dcn',
    Likes: '29',
    Username: 'ridafatema29',
    Comments: 'Excellent'
  },
  _id: ObjectId('69d62c01c19618fbaf44ba91')
}
]
RIDA>
```